NALAPRO Project:

Dataset:
from sklearn.datasets import **fetch_20newsgroups**

Task Description:

1. Train a simple neural network with two linear layers and a RELU in between.
a. do the required preprocessing on your data.
b. Train a word2vec network to generate the embedding vectors which are the inputs to your neural network. Observe and visualize the vectors after training your model for several epochs compared to training it with only one epoch. Did the vectors change in the space? Compare them together. What was the effect of the training?
c. Use TF-IDF as the vectorization of your inputs to the neural network. Compare the results to the results of b. What are the differences?

d. come up with one more experiment which could potentially improve your results while the network stays the same. Run this experiment and discuss the results.

2. Fine-tune a BERT model (base) on the training data and do evaluations and comparisons to the previous solution. What are the parameters and potential experiments you could run in this exercise. Present your results and discuss and evaluate them.

3. Fine-tune a BERT model by masking some words out instead of fine tuning the classification task. After that, now finetune the model on the classification task. Evaluate and discuss the results.

4. Using Llama-3 run zero-shot and few-shot learning experiments. Evaluate and discuss the results.

Deliverables:

1. Each student needs to create a repository on GitHub and make it accessible to the lecturer for the evaluations.
2. Each student needs to submit a report for the project. The report needs to written as a scientific report. Do not include any code there. Make sure you discuss the results, include your graphs and plots, comapred them to each other and include arguments on why the results are as they are.
3. The repository includes all your codes, documentations and report. Do not upload the dataset on your repository. Send the lecturer one email which includes the link to your repository and specify where your final report is on this repository.

4. Do not submit any codes per email, otherwise those codes and documents are not going to be evaluated.
5. The code
    a. needs to be well documented.
    b. There should be a section at the beginning where you provide the weights and biases (or Mlflow) link to your code as well as mentioning the tools you have used.
    c. You are allowed to use AI but you need to mention it under the tools you have used. And you must understand it and be able to explain it.
    d. If you use the code from any other sources, you must cite the source code.
    e. You are not allowed to copy code from other team students.
6. The deadline to submit your deliverables is EOD 06.06.2026.
7. The grading is as follows: 60% project work. 40% final exam.
   Project work: 30% the code. 30% report. 40% presentation
8. You will have a presentation on the last day of the lecture. In this presentation you have 15 min to present and 15 minutes to answer the questions.